

Acoustics Homework #3

Due April 1, 2026

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Note: I did this on paper and I transferred it to this notebook for ease in combining it with the plots needed.

Problem 1

Problem #1

▷ For a forced, driven SHO show that $Z_m = |\tilde{Z}_m|$ can be written in terms of the quality factor Q , frequency ω , and res. freq. ω_0 as,

$$Z_m(\omega) = R_n \left[1 + Q^2 \left(\frac{\omega}{\omega_0} \right)^2 \left(1 - \left(\frac{\omega}{\omega_0} \right)^2 \right)^2 \right]^{1/2}$$

and that the phase of $\tilde{Z}_m = Z_m e^{j\theta}$ can be written as,

$$\theta(\omega) = \tan^{-1} \left[Q \frac{\omega}{\omega_0} \left(1 - \frac{\omega_0^2}{\omega^2} \right) \right]$$

$$\rightarrow \tilde{Z}_m = R_n + j \left(\omega m - \frac{k_s}{\omega} \right) \rightarrow \omega_0 = \sqrt{\frac{k_s}{m}} \rightarrow k_s = m \omega_0^2$$

$$Q = \frac{\omega_0 m}{R_n} \rightarrow m = \frac{Q R_n}{\omega_0}$$

$$\rightarrow \text{Substitute } k_s = m \omega_0^2 \text{ into reactance term } (X_n) \rightarrow X_n = \omega m - \frac{m \omega_0^2}{\omega} = m \left(\omega - \frac{\omega_0^2}{\omega} \right) = m \omega \left(1 - \frac{\omega_0^2}{\omega^2} \right)$$

$$\rightarrow \text{Substitute } m = \frac{Q R_n}{\omega_0} \text{ in } \rightarrow X_n = \left(\frac{Q R_n}{\omega_0} \right) \omega \left(1 - \frac{\omega_0^2}{\omega^2} \right) = R_n Q \left(\frac{\omega}{\omega_0} \right) \left(1 - \left(\frac{\omega_0}{\omega} \right)^2 \right)$$

$$\text{Remember that } Z_m = |\tilde{Z}_m| = \sqrt{R_n^2 + X_n^2}$$

$$\rightarrow Z_m = \sqrt{R_n^2 + \left[R_n Q \left(\frac{\omega}{\omega_0} \right) \left(1 - \left(\frac{\omega_0}{\omega} \right)^2 \right) \right]^2} \rightarrow \text{factor out } R_n^2$$

$$\rightarrow \left[Z_m(\omega) = R_n \left[1 + Q^2 \left(\frac{\omega}{\omega_0} \right)^2 \left(1 - \left(\frac{\omega_0}{\omega} \right)^2 \right)^2 \right]^{1/2} \right]$$

For $\theta(\omega)$ the phase of a complex # $A + jB$ is given as $\tan^{-1} \left(\frac{B}{A} \right)$

$$\rightarrow \theta(\omega) = \tan^{-1} \left(\frac{X_n}{R_n} \right) \rightarrow \theta(\omega) = \tan^{-1} \left[\frac{R_n Q \left(\frac{\omega}{\omega_0} \right) \left(1 - \left(\frac{\omega_0}{\omega} \right)^2 \right)}{R_n} \right]$$

$$\text{This leaves us with } \rightarrow \theta(\omega) = \tan^{-1} \left[Q \frac{\omega}{\omega_0} \left(1 - \frac{\omega_0^2}{\omega^2} \right) \right]$$

Plots below

In [2]: # Plots for Problem 1

```
import numpy as np
import matplotlib.pyplot as plt
```

```
# Define the range for w_ratio (which is w/w0)
```

```
w_ratio = np.logspace(-1, 1, 500) # 0.1 to 10 on a log scale for better visualizati
```

```
# Define the Q values
```

```
Q_values = [1, 10]
```

```
# Setup the figure
```

```
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(8, 10))
```

```
for Q in Q_values:
```

```

# Normalized Magnitude:  $Z_m / R_m$ 
Z_norm = np.sqrt(1 + (Q**2) * (w_ratio**2) * (1 - (1 / w_ratio)**2)**2)

# Phase in radians
Phase_rad = np.arctan(Q * w_ratio * (1 - (1 / w_ratio)**2))

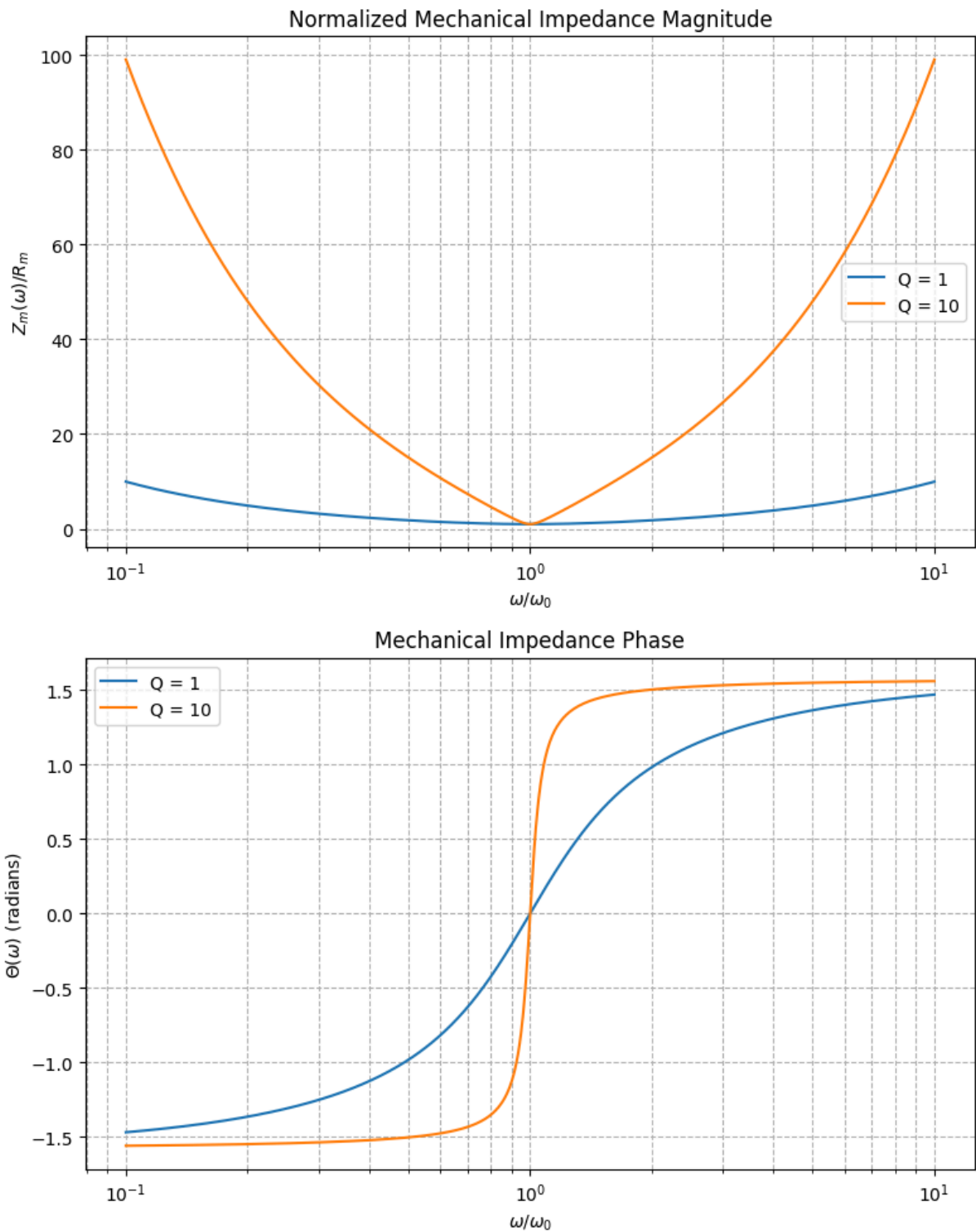
# Plotting
ax1.semilogx(w_ratio, Z_norm, label=f'Q = {Q}')
ax2.semilogx(w_ratio, Phase_rad, label=f'Q = {Q}')

# Format Magnitude Plot
ax1.set_title('Normalized Mechanical Impedance Magnitude')
ax1.set_ylabel(r'$Z_m(\omega) / R_m$')
ax1.set_xlabel(r'$\omega / \omega_0$')
ax1.grid(True, which="both", ls="--")
ax1.legend()

# Format Phase Plot
ax2.set_title('Mechanical Impedance Phase')
ax2.set_ylabel(r'$\Theta(\omega)$ (radians)')
ax2.set_xlabel(r'$\omega / \omega_0$')
ax2.grid(True, which="both", ls="--")
ax2.legend()

plt.tight_layout()
plt.show()

```



Problem 2

This one is mainly just plots. My plot of a waveform that moves right with a constant speed without changing shape is down below.

In [3]: `# Plots for Problem 2`

```
import numpy as np
```

```

import matplotlib.pyplot as plt

# Define wave parameters
A = 1.0          # Amplitude
omega = np.pi   # Angular frequency
k = np.pi       # Wave number (yields a speed  $v = \omega/k = 1.0$  m/s)

# Define position array
x = np.linspace(0, 5, 500)

# Define a few time steps to show movement
times = [0.0, 0.5, 1.0]

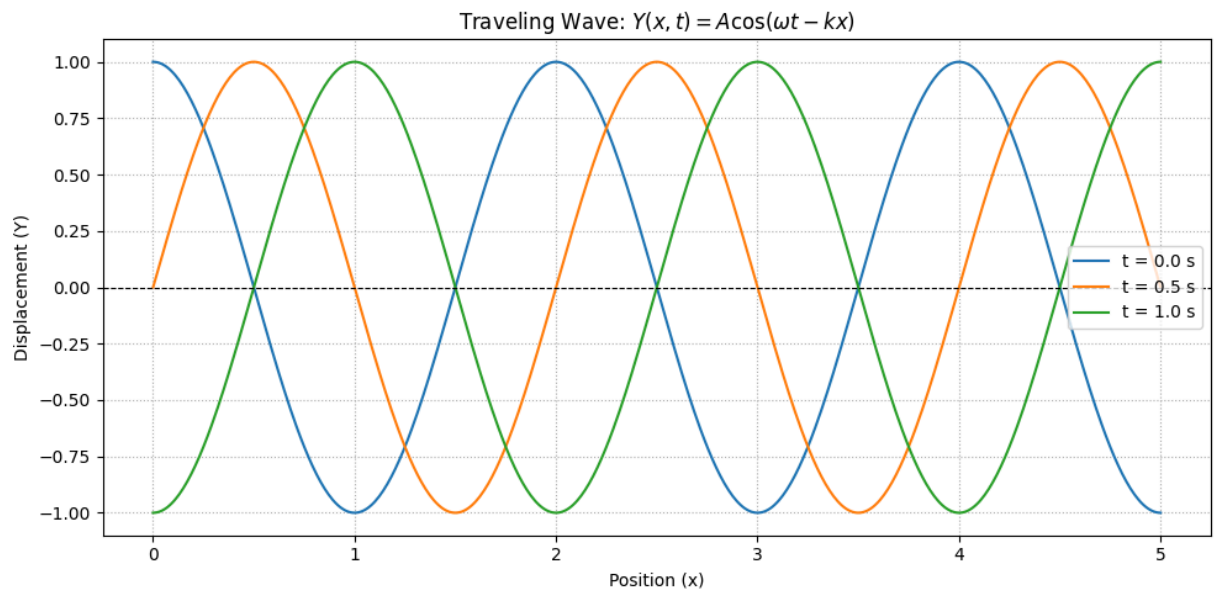
# Set up the plot
plt.figure(figsize=(10, 5))

for t in times:
    # Calculate wave at time t
    y = A * np.cos(omega * t - k * x)
    plt.plot(x, y, label=f't = {t} s')

# Formatting
plt.title(r'Traveling Wave:  $Y(x,t) = A \cos(\omega t - kx)$ ')
plt.xlabel('Position (x)')
plt.ylabel('Displacement (Y)')
plt.axhline(0, color='black', linewidth=0.8, linestyle='--')
plt.legend()
plt.grid(True, linestyle=':')

plt.tight_layout()
plt.show()

```



Problem 3

No plots needed.

Standard wave eqn.: $y(x,t) = A \cos(\omega t - kx + \phi)$

$$y(x,t) = 0.05 \cos(4t - 3x)$$

$A \rightarrow$ Amplitude $\rightarrow 0.05 \text{ m}$

$\omega \rightarrow$ Ang freq. $\rightarrow 4 \text{ rad/s}$

$k \rightarrow$ Wave # $\rightarrow 3 \text{ rad/m}$

$\rho_L \rightarrow$ linear mass density $\rightarrow 1.00 \times 10^{-4} \text{ kg/m}$

c.) Tension? $\rightarrow$ ~~$T = \rho_L v^2$~~

$$v = \sqrt{\frac{T}{\rho_L}} \rightarrow T = \rho_L v^2$$

$$T = (1.00 \times 10^{-4} \text{ kg/m})(1.33 \text{ m/s})^2$$

$$T = 1.78 \times 10^{-4} \text{ N}$$

a.) Peak to peak $\rightarrow 2A$

$$\rightarrow 2(0.05) = [0.1 \text{ m}]$$

b.) Transverse wave speed v

$$v = \frac{\omega}{k} = \frac{4}{3} \approx 1.33 \text{ m/s}$$

d.) λ wavelength?

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{3 \text{ rad/m}} \approx 2.09 \text{ m}$$

e.) Transvers string speed? Wave speed is how fast energy moves horizontally.

String speed is how fast the string moves up & down. To find this, take derivative of displacement (y) with respect to time (t).

$$\begin{aligned} \rightarrow v_y(x,t) &= \frac{dy}{dt} = -0.05(4) \sin(4t - 3x) \\ &= -0.2 \sin(4t - 3x) \end{aligned}$$

The max value of the sine function is 1

$$\begin{aligned} \text{So the max speed is } v_{y,\text{max}} &= A\omega = (0.05 \text{ m})(4 \text{ rad/s}) \\ &= [0.2 \text{ m/s}] \end{aligned}$$

Problem #3

Acoustics HW #3

Caleb Young

Page 3

Problem 4

Plots below

$$\frac{\partial y}{\partial t} = \sum_{m=1}^{\infty} C_m \omega_m \sin\left(\frac{m\pi x}{L}\right) \cos(\omega_m t) \rightarrow t=0 \rightarrow \cos(0)=1$$

Problem #4

- ▷ Calculate values of C_m for a string excited by the following triangular Velocity distribution applied to one-half of the string's length.

$$V_0(x,0) = \begin{cases} \frac{V_1 x}{L/2}, & 0 < x < \frac{L}{2} \\ \frac{V_1}{L} \left(\frac{L}{2} - x\right), & \frac{L}{2} < x < L \\ 0, & x > L \end{cases}$$

You'll find: $C_m = \frac{8V_1 \sin^2\left(\frac{m\pi}{8}\right) \sin\left(\frac{m\pi}{4}\right)}{\omega_m m^2 \pi^2}$. To confirm use below expression for C_m in expression of velocity at $t=0$

Assume that $V_1 = 1 \text{ m/s}$, $L = 1 \text{ m}$, and plot $V_0(x)$ for N_{max} equals 10, 30, 60, and 110 terms

$$y(x,0) = V_0(x) = \sum_{m=1}^{N_{\text{max}}} C_m \sin\left(\frac{m\pi x}{L}\right)$$

- ▷ General sol. for displacement of vibrating string w fixed ends is:

$$\rightarrow y(x,t) = \sum_{m=1}^{\infty} \sin\left(\frac{m\pi x}{L}\right) [A_m \cos(\omega_m t) + C_m \sin(\omega_m t)] \rightarrow \text{Since there's no initial displ. } (y(x,0)=0), A_m=0. \text{ To use this take partial der. w respect to time } \rightarrow \frac{\partial y}{\partial t}$$

→ at $t=0$ this becomes a Fourier sine series w coefficient $(C_m \omega_m)$

$$\rightarrow C_m = \frac{2}{L \omega_m} \int_0^L V_0(x,0) \sin\left(\frac{m\pi x}{L}\right) dx \rightarrow \text{Piecewise function must split, } x > \frac{L}{2} \text{ is 0}$$

$$\text{Let } k = \frac{m\pi}{L} \rightarrow C_m = \frac{2}{L \omega_m} \left[\int_0^{L/2} \frac{V_1 x}{L} \sin(kx) dx + \int_{L/2}^L \frac{V_1}{L} \left(\frac{L}{2} - x\right) \sin(kx) dx \right]$$

$$\rightarrow 1^{\text{st}} \text{ integral} \rightarrow C_{m1} = \frac{V_1}{L k^2} \sin\left(\frac{m\pi}{4}\right) - \frac{V_1}{4k} \cos\left(\frac{m\pi}{4}\right)$$

$$\rightarrow 2^{\text{nd}} \text{ integral} \rightarrow C_{m2} = \frac{V_1}{2} \left[-\frac{\cos(kx)}{k} \right]_{L/2}^L - \frac{V_1}{L} \left[\frac{\sin(kx)}{k} - \frac{x \cos(kx)}{k} \right]_{L/2}^L$$

$$C_{m2} = -\frac{V_1}{2k} \cos\left(\frac{m\pi}{2}\right) + \frac{V_1}{2k} \cos\left(\frac{m\pi}{4}\right) - \frac{V_1}{Lk^2} \sin\left(\frac{m\pi}{2}\right) + \frac{V_1}{2k} \cos\left(\frac{m\pi}{2}\right) + \frac{V_1}{Lk^2} \sin\left(\frac{m\pi}{4}\right) - \frac{V_1}{4k} \cos\left(\frac{m\pi}{4}\right)$$

This reminds me of Cal 2 → Combine → $C_m = \frac{V_1}{Lk^2} \left[2\sin\left(\frac{m\pi}{4}\right) - \sin\left(\frac{m\pi}{2}\right) \right]$

$$\rightarrow k^2 = \frac{m^2 \pi^2}{L^2} \text{ ; Sub in } \frac{2}{L \omega_m} \rightarrow C_m = \frac{2V_1}{\omega_m m^2 \pi^2} \left[2\sin\left(\frac{m\pi}{4}\right) - \sin\left(\frac{m\pi}{2}\right) \right]$$

→ After applying the double angle identity for sine to the 2nd term we have our result

$$\rightarrow C_m = \frac{2V_1}{\omega_m m^2 \pi^2} \left[\sin\left(\frac{m\pi}{4}\right) \sin\left(\frac{m\pi}{8}\right) \right] \rightarrow \left[C_m = \frac{8V_1 \sin^2\left(\frac{m\pi}{8}\right) \sin\left(\frac{m\pi}{4}\right)}{\omega_m m^2 \pi^2} \right]$$

In [5]: # Plots for Problem 4

```
import numpy as np
import matplotlib.pyplot as plt
```

```
# Given constants
```

```
v1 = 1.0 # m/s
```

```
L = 1.0 # m
```

```
n_max_values = [10, 30, 60, 110]
```

```

# Define x range across the whole string
x = np.linspace(0, L, 1000)

# 1. Define the 'True' analytical piecewise function for comparison
def true_v0(x_val):
    if 0 < x_val < L/4:
        return (v1 * x_val) / L
    elif L/4 <= x_val < L/2:
        return (v1 / L) * (L/2 - x_val)
    else:
        return 0

v0_analytical = np.array([true_v0(val) for val in x])

# 2. Setup the plot
plt.figure(figsize=(12, 8))
plt.plot(x, v0_analytical, 'k--', linewidth=2, label='Analytical (True) v0(x)', alp

# 3. Calculate and plot the Fourier Series for each N_max
for N_max in n_max_values:
    v0_reconstructed = np.zeros_like(x)

    for m in range(1, N_max + 1):
        # The sum uses omega_m * C_m.
        # From the problem: omega_m * C_m = [8 * v1 * sin^2(m*pi/8) * sin(m*pi/4)]
        coeff = (8 * v1 * (np.sin(m * np.pi / 8)**2) * np.sin(m * np.pi / 4)) / (m*

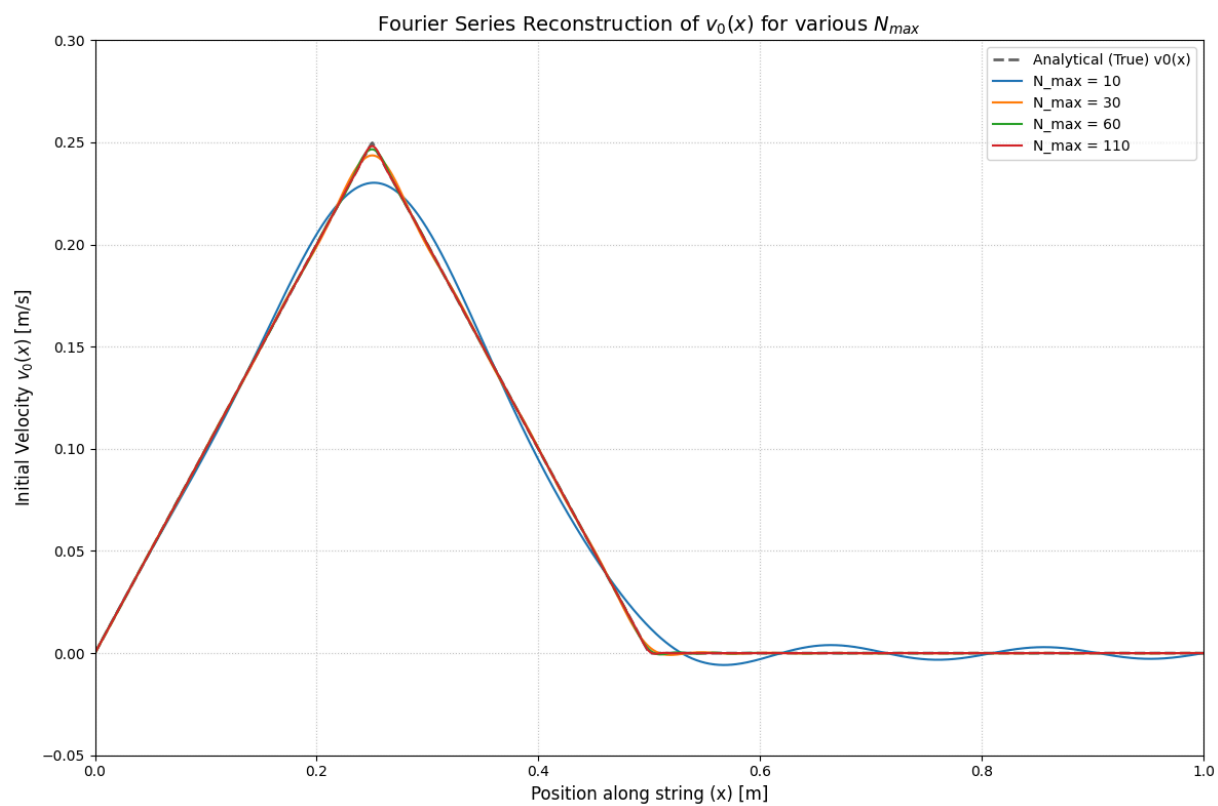
        # Add the m-th harmonic to the sum
        v0_reconstructed += coeff * np.sin(m * np.pi * x / L)

    plt.plot(x, v0_reconstructed, label=f'N_max = {N_max}')

# Formatting
plt.title(r'Fourier Series Reconstruction of $v_0(x)$ for various $N_{\max}$', fonts
plt.xlabel('Position along string (x) [m]', fontsize=12)
plt.ylabel('Initial Velocity $v_0(x)$ [m/s]', fontsize=12)
plt.legend(loc='upper right')
plt.grid(True, linestyle=':', alpha=0.7)
plt.xlim(0, L)
plt.ylim(-0.05, 0.3) # Height is v1/4 = 0.25

plt.tight_layout()
plt.show()

```

Problem 5

Problem #5

Prove that energy is conserved for 2 strings w different (real) impedances attached by a massless knot at the origin using the power reflection and power transmission coefficients, i.e. show $\mathcal{T}_R + \mathcal{T}_T = 1$

▷ We need to express reflection & transmission in terms of impedances.
 $Z_1 \& Z_2 \rightarrow R = \frac{Z_1 - Z_2}{Z_1 + Z_2} \rightarrow T = \frac{2Z_1}{Z_1 + Z_2}$

$$\mathcal{T}_R = \frac{P_{\text{reflected}}}{P_{\text{incident}}} = \frac{Z_1 R^2}{Z_1 (1)^2} = R^2 \Rightarrow R^2 = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2} = \mathcal{T}_R$$

$$\mathcal{T}_T = \frac{Z_2 T^2}{Z_1 (1)^2} = \frac{Z_2}{Z_1} T^2 \rightarrow \text{sub} \rightarrow \mathcal{T}_T = \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2}$$

(Common denominator)
 $(Z_1 - Z_2)^2 = Z_1^2 - 2Z_1 Z_2 + Z_2^2$

$$\mathcal{T}_R + \mathcal{T}_T = \frac{Z_1^2 - 2Z_1 Z_2 + Z_2^2 + 4Z_1 Z_2}{(Z_1 + Z_2)^2} \rightarrow \frac{Z_1^2 + Z_2^2 + 2Z_1 Z_2}{(Z_1 + Z_2)^2}$$

$$\rightarrow \frac{Z_1^2 + 2Z_1 Z_2 + Z_2^2}{(Z_1 + Z_2)^2} = \frac{(Z_1 + Z_2)^2}{(Z_1 + Z_2)^2} = 1$$

So $\mathcal{T}_R + \mathcal{T}_T = 1$, energy is conserved